Budget-Robust Station Selection for Microtransit Under Demand Uncertainty

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Abstract

We extend a two-stage stochastic virtual bus stop (VBS) selection model to handle uncertain passenger demand within each scenario type (morning peak, midday, evening peak, weekend). Demand for each origin–destination pair (o, d) and scenario type s is modeled as lying in a budget uncertainty set parameterized by lower and upper bounds derived from historical data and a per-scenario budget Γ_s that limits the number of OD pairs allowed to simultaneously reach their upper demand levels. We derive the dual robust counterpart of the inner maximization, which is a linear program with the same structure as the nominal model. We compare three station-selection policies— *Nominal*, *Upper-Bound*, and *Budget-Robust*—on out-of-sample demand realizations and report average cost, high-percentile (P90/P95) cost, and the price of robustness.

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1 Introduction

Microtransit systems offer flexible, on-demand shared rides between a set of designated virtual bus stops (VBS). A key design decision is which candidate locations to build as VBS. Prior work formulates this as a two-stage stochastic mixed-integer program: in the first stage, build l stations and choose which $k \leq l$ are active in each scenario $s \in S$; in the second stage, assign passengers to origin–destination VBS pairs.

A limitation of the stochastic model is that it treats the demand q_{ods} within each scenario as a fixed, known quantity (typically the historical mean). In practice, demand fluctuates even within a scenario type: morning-peak demand on one day may differ substantially from the next. This motivates a *robust* extension where q_{ods} is uncertain within a bounded set.

We adopt Bertsimas and Sim's *budget uncertainty set* [1], which gives a tractable linear robust counterpart and allows a meaningful trade-off between conservatism and solution quality.

2 Base Model

2.1 Decision variables

$y_j \in \{0,1\}$	build station j
$z_{js} \in \{0,1\}$	station j is active in scenario s
$x_{odjks} \in \{0,1\}$	assign OD pair (o,d) to origin VBS j, destination VBS k in scenario s

2.2 Nominal objective

The nominal two-stage objective minimizes expected passenger travel cost:

$$\min_{x,y,z} \sum_{s\in S} \sum_{(o,d)\in\Omega} \sum_{j,k\in J} q_{ods} \underbrace{\left(d_{oj}^{\text{orig}} + d_{dk}^{\text{dest}} + \lambda\, c_{jk}\right)}_{a_{odjks}} x_{odjks}, \tag{1}$$

where d_{oj}^{orig} is the walking distance from origin o to VBS j, d_{dk}^{dest} is the walking distance from VBS k to destination d, c_{jk} is the in-vehicle travel time between VBS j and k, and λ is a weight that converts routing time to the same unit as walking distance.

2.3 Constraints

$$\sum_{j,k} x_{odjks} = 1 \qquad \forall (o,d), s \tag{2}$$

$$x_{odjks} \le z_{js}, \quad x_{odjks} \le z_{ks} \qquad \forall (o,d), j, k, s \tag{3}$$

$$z_{js} \le y_j \qquad \forall j, s \tag{4}$$

$$\sum_j y_j = l, \quad \sum_j z_{js} = k \qquad \forall s \tag{5}$$

3 Budget Uncertainty Set for Demand

3.1 Parameterization

For each OD pair (o,d) and scenario type s, assume we can estimate a lower bound $\underline{q}_{ods}$ and an upper bound $\overline{q}_{ods}$ on demand. Write

$$q_{ods} = \bar{q}_{ods} + \hat{q}_{ods}\,\xi_{ods}, \qquad \bar{q}_{ods} = \underline{q}_{ods}, \qquad \hat{q}_{ods} = \overline{q}_{ods} - \underline{q}_{ods}, \qquad 0 \le \xi_{ods} \le 1. \tag{6}$$

Impose a *budget constraint* per scenario type:

$$\sum_{(o,d)\in\Omega} \xi_{ods} \le \Gamma_s \qquad \forall s \in S. \tag{7}$$

The uncertainty set is then

$$\mathcal{U}_s = \Big\{ q_s : q_{ods} = \underline{q}_{ods} + \hat{q}_{ods}\,\xi_{ods},\ 0 \le \xi_{ods} \le 1,\ \textstyle\sum_{(o,d)} \xi_{ods} \le \Gamma_s \Big\}. \tag{8}$$

This set is best interpreted as a budget on *normalized upward deviations* from the lower-bound demand vector, not as a direct statement about integer passenger counts. The variable ξ_{ods} measures the fraction of the interval $[\underline{q}_{ods}, \overline{q}_{ods}]$ that is activated for OD pair (o,d) in scenario s.

3.2 Interpretation of Γ_s

- $\Gamma_s = 0$: all OD pairs at their lower bound (most optimistic).
- $\Gamma_s = |\Omega|$: all OD pairs can simultaneously reach their upper bound (equivalent to the upper-bound conservative model).
- Intermediate Γ_s: guards against a total normalized deviation of at most Γ_s across OD pairs.

This is not, in general, the same as saying that at most Γ_s OD pairs spike. That counting interpretation becomes exact only when the inner maximizer chooses $\xi_{ods} \in \{0,1\}$, which happens for the continuous-knapsack worst-case problem when Γ_s is an integer and all induced costs are nonnegative. If Γ_s is fractional, one OD pair may be partially activated in the worst case.

A larger Γ_s for morning peak than for midday encodes the prior that peak hours exhibit more correlated demand spikes.

3.3 Remark on integer demand

Observed demand is integer-valued, but the robust inner maximization is written over continuous variables ξ_{ods}. This is not automatically a relaxation gap. For a fixed first-stage solution x, the inner problem is

$$\max_{\xi} \sum_{(o,d)} \hat{q}_{ods} C_{ods}(x)\, \xi_{ods} \quad \text{s.t. } 0 \le \xi_{ods} \le 1,\ \sum_{(o,d)} \xi_{ods} \le \Gamma_s.$$

Since $C_{ods}(x) \ge 0$, this is a continuous knapsack problem. If $\Gamma_s \in \mathbb{Z}_+$, there exists an optimal solution with $\xi_{ods} \in \{0,1\}$: sort OD pairs by $\hat{q}_{ods} C_{ods}(x)$ and set the largest Γ_s entries to 1. Hence the worst-case objective value is exact even though the formulation uses continuous ξ. The caveat is interpretive: extreme points correspond to lower-bound or upper-bound demand levels, not arbitrary intermediate integer counts inside $[\underline{q}_{ods}, \overline{q}_{ods}]$.

3.4 Calibrating bounds and budget from data

Group historical observations by scenario type $s \in \{\text{morning}, \text{midday}, \text{evening}, \text{weekend}\}$. For each OD pair (o,d) compute empirical quantiles:

$$\underline{q}_{ods} = Q_{0.25}\big(q_{od}^{\text{hist}} \mid s\big), \qquad \overline{q}_{ods} = Q_{0.90}\big(q_{od}^{\text{hist}} \mid s\big). \tag{9}$$

Then define the normalized demand inflation for each historical day:

$$\xi_{ods}^{\text{hist}} = \text{clip}\left(\frac{q_{ods}^{\text{hist}} - \underline{q}_{ods}}{\hat{q}_{ods}},\ 0,\ 1 \right). \tag{10}$$

Compute the total budget consumed on each historical day:

$$B_s^{\text{hist}} = \sum_{(o,d)\in\Omega} \xi_{ods}^{\text{hist}}. \tag{11}$$

Choose

$$\Gamma_s = Q_{0.90}\big(B_s^{\text{hist}}\big) \quad \text{or} \quad \Gamma_s = Q_{0.95}\big(B_s^{\text{hist}}\big). \tag{12}$$

3.5 Additional uncertainty set based on total demand

In addition to the budgeted-deviation set (8), one can also study a total-demand-cap uncertainty set if the modeling intent is not to limit the amount of perturbation, but rather to impose an upper bound on total demand:

$$\mathcal{U}_s^{\text{sum}} = \Big\{ q_s : \underline{q}_{ods} \le q_{ods} \le \overline{q}_{ods}\ \forall (o,d),\ \sum_{(o,d)\in\Omega} q_{ods} \le \overline{Q}_s \Big\}. \tag{13}$$

This added uncertainty set is not equivalent to the budget set (8) in general. Substituting $q_{ods} = \underline{q}_{ods} + \hat{q}_{ods}\xi_{ods}$ yields

$$\sum_{(o,d)} \hat{q}_{ods}\, \xi_{ods} \le \overline{Q}_s - \sum_{(o,d)} \underline{q}_{ods}, \tag{14}$$

which is a *weighted* budget on deviations. By contrast, (8) uses the unweighted budget $\sum \xi_{ods} \le \Gamma_s$. These two descriptions are equivalent only in the special case where all deviation widths $\hat{q}_{ods}$ are identical.

The inner worst-case problem under (13) remains tractable:

$$\max_{q} \sum_{(o,d)} q_{ods} C_{ods}(x) \quad \text{s.t. } \underline{q}_{ods} \le q_{ods} \le \overline{q}_{ods}, \ \sum_{(o,d)} q_{ods} \le \overline{Q}_s.$$

If all bounds and $\overline{Q}_s$ are integral, the LP relaxation is exact, so integrality may be dropped in the inner problem without changing the worst-case value. This makes the total-demand-cap model a useful extension to the current discussion when the desired policy statement is a cap on $\sum q_{ods}$ rather than a cap on aggregate normalized deviation.

3.6 Robust counterpart for the total-demand-cap set

Assume

$$\sum_{(o,d)} \underline{q}_{ods} \le \overline{Q}_s \le \sum_{(o,d)} \overline{q}_{ods} \qquad \forall s \in S,$$

so that the uncertainty set (13) is nonempty. Define the remaining demand budget after accounting for lower bounds:

$$B_s := \overline{Q}_s - \sum_{(o,d)} \underline{q}_{ods}.$$

Now substitute

$$q_{ods} = \underline{q}_{ods} + r_{ods}, \qquad 0 \le r_{ods} \le \hat{q}_{ods}.$$

Then, for fixed x, the inner worst-case problem becomes

$$\max_{r} \sum_{(o,d)} \big[\underline{q}_{ods} + r_{ods}\big] C_{ods}(x) \quad \text{s.t. } 0 \le r_{ods} \le \hat{q}_{ods}, \ \sum_{(o,d)} r_{ods} \le B_s. \tag{15}$$

Separating the constant term gives

$$\sum_{(o,d)} \underline{q}_{ods} C_{ods}(x) + \max_{r} \left\{ \sum_{(o,d)} r_{ods} C_{ods}(x) : 0 \le r_{ods} \le \hat{q}_{ods}, \ \sum_{(o,d)} r_{ods} \le B_s \right\}.$$

The second term is again a continuous knapsack problem, now with upper bounds $\hat{q}_{ods}$ on the excess-demand variables r_{ods}.

Proposition 1. For each scenario s, the inner maximization (15) has dual

$$\begin{aligned} &\max_{r} \sum_{(o,d)} r_{ods} C_{ods}(x) && \text{s.t. } 0 \le r_{ods} \le \hat{q}_{ods}, \ \sum_{(o,d)} r_{ods} \le B_s \\ = &\min_{\alpha_s \ge 0,\, \beta_{ods} \ge 0} B_s\, \alpha_s + \sum_{(o,d)} \hat{q}_{ods}\, \beta_{ods} && \text{s.t. } \alpha_s + \beta_{ods} \ge C_{ods}(x)\ \forall (o,d). \end{aligned} \tag{16}$$

Therefore the robust counterpart under the total-demand-cap uncertainty set is

$$\min_{x,y,z,\alpha,\beta} \sum_{s} \left[\sum_{(o,d)} \underline{q}_{ods}\, C_{ods}(x) + B_s\, \alpha_s + \sum_{(o,d)} \hat{q}_{ods}\, \beta_{ods} \right] \tag{17}$$

subject to

$$\alpha_s + \beta_{ods} \ge C_{ods}(x) \qquad \forall (o,d),\ s, \tag{18}$$

plus the original constraints (2)–(5). Expanding $C_{ods}(x) = \sum_{j,k} a_{odjks}\, x_{odjks}$ gives the fully explicit linear form:

$$\min_{x,y,z,\alpha,\beta} \sum_{s} \sum_{(o,d)} \sum_{j,k} \underline{q}_{ods}\, a_{odjks}\, x_{odjks} + \sum_{s} B_s\, \alpha_s + \sum_{s} \sum_{(o,d)} \hat{q}_{ods}\, \beta_{ods} \tag{19}$$

subject to

$$\alpha_s + \beta_{ods} \geq \sum_{j,k} a_{odjks}\, x_{odjks} \qquad \forall (o,d),\ s, \tag{20}$$

plus (2)–(5).

Remark 1. Compared with the Bertsimas-Sim counterpart (24), the roles of the deviation widths move from the right-hand side of the robust constraints to the objective coefficients of the dual variables β_{ods}. This is the key structural difference induced by placing the budget on total demand $\sum q_{ods}$ instead of on normalized deviations.

4 Robust Objective

The robust model replaces the nominal objective with

$$\min_{x,y,z} \sum_{s \in S} \max_{q_s \in \mathcal{U}_s} \sum_{(o,d) \in \Omega} q_{ods}\, C_{ods}(x), \tag{21}$$

where the induced assignment cost $C_{ods}(x) = \sum_{j,k} a_{odjks}\, x_{odjks}$ depends only on the first-stage solution.

Substituting (6):

$$\max_{\xi} \sum_{(o,d)} \left[\underline{q}_{ods} + \hat{q}_{ods}\, \xi_{ods}\right] C_{ods}(x) \quad \text{s.t. } 0 \leq \xi_{ods} \leq 1,\ \textstyle\sum_{(o,d)} \xi_{ods} \leq \Gamma_s. \tag{22}$$

This decomposes into a *continuous knapsack* problem in ξ, which has a closed-form greedy solution but is most conveniently dualized.

5 Dual Robust Counterpart

Proposition 2. The inner maximization (22) has the following dual:

$$\begin{aligned} &\max_{\xi} \sum_{(o,d)} \hat{q}_{ods}\, C_{ods}(x)\, \xi_{ods} && \text{s.t. } 0 \leq \xi_{ods} \leq 1,\ \sum_{(o,d)} \xi_{ods} \leq \Gamma_s \\ = &\min_{\eta_s \geq 0,\, \rho_{ods} \geq 0} \Gamma_s\, \eta_s + \sum_{(o,d)} \rho_{ods} && \text{s.t. } \eta_s + \rho_{ods} \geq \hat{q}_{ods}\, C_{ods}(x)\ \forall (o,d). \end{aligned} \tag{23}$$

Substituting back into (21), the full robust counterpart is:

$$\min_{x,y,z,\eta,\rho} \sum_{s} \sum_{(o,d)} \sum_{j,k} \underline{q}_{ods}\, a_{odjks}\, x_{odjks} + \sum_{s} \Gamma_s\, \eta_s + \sum_{s} \sum_{(o,d)} \rho_{ods} \tag{24}$$

subject to

$$\eta_s + \rho_{ods} \geq \hat{q}_{ods} \sum_{j,k} a_{odjks}\, x_{odjks} \qquad \forall (o,d),\ s, \tag{25}$$

plus the original constraints (2)–(5).

Remark 2. Constraint (25) is already linear. Indeed, $\hat{q}_{ods}$ is a parameter and $C_{ods}(x) = \sum_{j,k} a_{odjks}\, x_{odjks}$ is linear in x, so expanding (25) yields a standard linear inequality.

6 JuMP Implementation Sketch

The new data requirements are:

```
q_low[od, s]       # lower demand bound  (q_low_ods)
q_high[od, s]      # upper demand bound  (q_high_ods)
q_hat[od, s]       # = q_high - q_low   (q_hat_ods)
Gamma[s]           # budget of uncertainty
a[od, j, k, s]     # assignment cost
```

New variables:

```
@variable(model, eta[s in S] >= 0)
@variable(model, rho[od in OD, s in S] >= 0)
```

Robust objective:

```
@objective(model, Min,
    sum(q_low[od,s] * a[od,j,k,s] * x[od,j,k,s]
        for s in S, od in OD, j in J, k in J)
    + sum(Gamma[s] * eta[s] for s in S)
    + sum(rho[od,s] for od in OD, s in S)
)
```

Robust dual constraints:

```
@constraint(model, [od in OD, s in S],
    eta[s] + rho[od,s] >=
        q_hat[od,s] *
        sum(a[od,j,k,s] * x[od,j,k,s] for j in J, k in J)
)
```

All original constraints (2)–(5) are unchanged.

7 Evaluation Experiment

7.1 Three model variants

We compare:

1. **Nominal model.** Uses $q_{ods} = \mathbb{E}[q_{ods}]$ (scenario-average historical demand).
2. **Upper-bound model.** Sets $q_{ods} = \overline{q}_{ods}$ for every OD pair. Maximally conservative; equivalent to budget-robust with $\Gamma_s = |\Omega|$.
3. **Budget-robust model.** Uses uncertainty set $\mathcal{U}_s$ with Γ_s calibrated from data via (12).

Each model produces a station selection $(y^\star, z^\star)$ and assignments $(x^\star)$ that are then *fixed* for out-of-sample evaluation.

7.2 Out-of-sample evaluation protocol

For each held-out demand realization q^{test}_{ods} (from a held-out date range or a parametric simulation), compute the realized cost:

$$\text{Cost}(x^\star, q^{\text{test}}) = \sum_{s,(o,d),j,k} q^{\text{test}}_{ods}\, a_{odjks}\, x^\star_{odjks}. \tag{26}$$

7.3 Reported metrics

For each model, report over the test set:

- Mean realized cost
- 90th and 95th percentile realized cost
- Worst-case realized cost
- Number of OD pairs poorly assigned (realized walking distance exceeding a threshold)
- Stability of station selection across demand realizations
- *Price of robustness*: $\dfrac{\text{Robust obj} - \text{Nominal obj}}{\text{Nominal obj}}$

7.4 Expected finding

The budget-robust model protects against simultaneous OD demand spikes without assuming that every OD pair reaches its upper bound. It should give a *less conservative* solution than the full upper-bound model (lower in-sample objective, lower price of robustness) while improving high-percentile out-of-sample performance relative to the nominal model.

Formally:

$$\text{Nominal P90} \;\geq\; \text{Budget-Robust P90} \;\geq\; \text{Upper-Bound P90},$$

$$\text{Nominal mean} \;\leq\; \text{Budget-Robust mean} \;\leq\; \text{Upper-Bound mean}.$$

7.5 Experiment pipeline

The experiment is driven by scripts in `scripts/`:

1. `scripts/01_setup.jl`: reads `experiments/<name>/base.toml` and `experiments/<name>/sweep.toml`, generates per-job TOML configs.
2. `scripts/02_submit.sh`: SLURM array job template.
3. `scripts/03_run_job.jl`: loads config, runs the appropriate model, evaluates out-of-sample performance, saves `metrics.json`.
4. `scripts/aggregate.jl`: loads all `metrics.json` files, produces `analysis/results.csv` and prints summary statistics including the price of robustness.

The sweep in `experiments/2026-05-01_gamma_sweep/` varies $\Gamma_s \in \{0, 2, 5, 10, 20\}$ across all three model types and $k \in \{6, 8, 10\}$.

8 Conclusion

We propose a budget-robust extension of the VBS station selection model. The key insight is that the inner worst-case demand problem is a continuous knapsack with a clean dual, yielding a linear robust counterpart that can be solved with the same MILP solver as the nominal model. The budget Γ_s is a transparent, data-driven parameter that interpolates between the optimistic nominal model and the conservative upper-bound model. If the intended modeling assumption is instead a cap on total demand $\sum q_{ods}$, then the weighted uncertainty set (13)–(14) is the more direct formulation.

References

[1] D. Bertsimas and M. Sim, *The price of robustness*, Operations Research 52(1):35–53, 2004.

[2] D. Bertsimas and D. B. Brown, *Constructing uncertainty sets for robust linear optimization*, Operations Research 57(6):1483–1495, 2009.